

counterfeit, duplicate, illegal, smuggled, spurious and substandard agricultural inputs, especially pesticides or plant medicines. Fake seeds, duplicate fertilizers and adulterated pesticides have become a silent epidemic across much of the country. These products do not merely reduce yields; they destroy crops, erode farmer confidence and push families into debt. Recent raids in states such as **Uttar Pradesh, Madhya Pradesh, Gujarat, Haryana and Telangana** have uncovered factories producing and supplying **lakhs of bags of fake fertilizers every day** across state borders. The seriousness of the problem is evident from the decision of the **Chief Minister of Uttar Pradesh to invoke the National Security Act** against those involved in fertilizer counterfeiting and black marketing. This is not merely an economic offence causing losses of lakhs of crores to farmers—it is an attack on India's food security.

India already has strong laws such as the Seeds Act, the Fertilizer Control Order and the Insecticides Act, along with ongoing reforms like the proposed PMB 2025. The real challenge is not the absence of legislation but weak and inconsistent implementation. When enforcement is lax, illegal operators flourish, honest businesses suffer and farmers pay the price. What is urgently needed is strict enforcement supported by digital traceability, QR-code based verification, accredited laboratories and exemplary punishment for offenders.

Another structural challenge is farmer education and extension. With over **14 crore farmers across 6.5 lakh villages**, no government system, however well-intentioned, can reach everyone effectively on its own. This makes **public-private partnerships** essential. The private sector, with its innovation, field networks and digital platforms, must be incentivised to complement government efforts in research, training and technology delivery. When public institutions and private enterprises work together, farmers gain access to the best of science, markets and modern technologies.

Dhanuka Agritech brings cutting-edge technologies to Indian farmers through partnerships with **six leading Japanese companies**, enabling the introduction of **safer, greener and more efficient crop protection solutions** adapted for Indian conditions. While earlier products were applied in kilograms, today's advanced molecules are used in just a few grams, delivering higher efficacy with a much lower environmental footprint.

To strengthen indigenous innovation, Dhanuka has set up a **state-of-the-art R&D Centre at Palwal**, inaugurated by former Chief Minister **Shri Manohar Lal Khattar**. With **11 specialised laboratories**, it anchors Dhanuka's vision of making Palwal a **model agricultural district**. Through **DART**, the company provides **drone spraying, weather stations, soil and crop sensors, and precision advisory**, while also helping farmers adopt drone-based, technology-driven farming. In addition, a toll-free **farmer helpline (1800 102 1022)** is being run to provide real-time technical support and guidance.

In addition, Dhanuka has signed **MoUs with ICAR and 15 leading agricultural universities** across India, working closely with them on **research, technology validation and extension** to ensure that scientific innovations reach farmers' fields in a practical and scalable manner.

Despite these efforts, India's overall investment in research and development remains alarmingly low. R&D expenditure is only about **0.6% of GDP**, compared with nearly **6% in countries like Israel**. While the Hon'ble Prime Minister has given the inspiring call of "**Jai Anusandhan**," this vision cannot be realised without meaningful investment. Earlier, **DST-recognised companies** were allowed a **200% income-tax deduction on R&D spending**, which played a vital role in encouraging private innovation. The withdrawal of this incentive has weakened research spending at a time of intense global competition, and it therefore urgently needs to be **reinstated and further strengthened**.

India's national vision is bold and inspiring. The Prime Minister has set the target of a **USD 5 trillion economy**

by 2028, with agriculture contributing nearly USD 1 trillion, and a **USD 30 trillion Viksit Bharat by 2047**. These goals depend on transforming agriculture, as India's growth will pass through its farms, fields and villages. Agriculture must move from being subsidy-driven to one of **strength, productivity and enterprise**, where farmers earn more from every acre through better technology, assured quality inputs, efficient markets and value addition. The future lies in shifting from hand-holding to **self-reliance**, from giving fish to **providing the nets**. When farmers are trained, informed and protected from fraud, they no longer need perpetual subsidies; they become confident entrepreneurs and contributors to the national exchequer—making agriculture a true engine of India's economy. India has the land, the soil, the scientists, the industry, and the spirit. What we now need is a system that rewards honesty, innovation and farmer empowerment. If we get this right, Indian agriculture will not only feed the nation—it will drive India's rise as a global economic power.

